

Chapter

Linear In-equations In One Variable

1. The solution set for the inequation $2x + 4 \leq 14$, $x \in W$ is :

(a) $\{1, 2, 3, 4, 5\}$

(b) $\{0, 1, 2, 3, 4, 5\}$

(c) $\{1, 2, 3, 4\}$

(d) $\{0, 1, 2, 3, 4\}$

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2. Solve the following inequation. Write down the solution set and represent it on the real number line.

$$-5(x - 9) \geq 17 - 9x > x + 2, x \in R$$

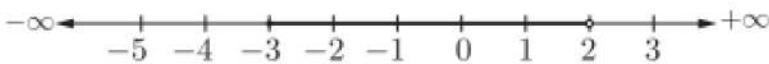
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3. Solve the following inequation, and graph the solution on the number line:

$$2x - 5 \leq 5x + 4 < 11, x \in R.$$

Comp 2023

4. The solution set representing the following number line is :



(a) $[x : x \in R, -3 \leq x < 2]$

(b) $[x : x \in R, -3 < x < 2]$

(c) $[x : x \in R, -3 \leq x \leq 2]$

(d) $[x : x \in R, -3 \leq x \leq 2]$

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5. The solution set of the inequation :

$x - 3 \leq -5, x \in R$ is :

- (a) $[x : x > -2, x \in R]$
- (b) $[x : x \leq -2, x \in R]$
- (c) $[x : x \geq -2, x \in R]$
- (d) $[-2, -1, 0, 1, 2]$

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6. The solution set on the number line of the linear inequation :

$2y - 6 < y + 2 \leq 2y, y \in N$ is :

- (a) 
- (b) 
- (c) 
- (d) 

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7. The solution set for the linear inequation

$-8 \leq x - 7 < -4, x \in I$ is :

- (a) $\{x : x \in R, -1 \leq x < 3\}$
- (b) $\{0, 1, 2, 3\}$
- (c) $\{-1, 0, 1, 2, 3\}$
- (d) $\{-1, 0, 1, 2\}$

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Solution

1.

Ans: Solve the inequation:

$$2x + 4 \leq 14$$

$$2x \leq 10$$

$$x \leq 5$$

Given $x \in W$ (whole numbers),

Whole numbers ≤ 5 are: $\{0, 1, 2, 3, 4, 5\}$

Correct answer: (b) $\{0, 1, 2, 3, 4, 5\}$

2.

Ans:

Given:

$$-5(x - 9) \geq 17 - 9x > x + 2, \quad x \in \mathbb{R}$$

Splitting into two inequalities:

$$-5(x - 9) \geq 17 - 9x \quad (1)$$

$$17 - 9x > x + 2 \quad (2)$$

Solving (1):

$$-5x + 45 \geq 17 - 9x$$

$$-5x + 9x \geq 17 - 45$$

$$4x \geq -28$$

$$x \geq -7 \quad (3)$$

Solving (2):

$$17 - 9x > x + 2$$

$$-9x - x > 2 - 17$$

$$-10x > -15$$

Multiplying by -1 (inequality sign reverses):

$$10x < 15$$

$$x < 1.5 \quad (4)$$

From (3) and (4):

$$-7 \leq x < 1.5, \quad x \in \mathbb{R}$$

Solution set:

$$\boxed{\{x : -7 \leq x < 1.5, x \in \mathbb{R}\}}$$

Number line representation:

Closed circle at -7 , open circle at 1.5 , shaded region between them.

3.

Ans:

Given:

$$2x - 5 \leq 5x + 4 < 11, \quad x \in \mathbb{R}$$

Splitting into two inequalities:

$$2x - 5 \leq 5x + 4 \quad (1)$$

$$5x + 4 < 11 \quad (2)$$

Solving (1):

$$2x - 5 \leq 5x + 4$$

$$2x - 5x \leq 4 + 5$$

$$-3x \leq 9$$

Dividing by -3 (inequality sign reverses):

$$x \geq -3 \quad (3)$$

Solving (2):

$$5x + 4 < 11$$

$$5x < 7$$

$$x < \frac{7}{5} \quad (4)$$

From (3) and (4):

$$-3 \leq x < \frac{7}{5}, \quad x \in \mathbb{R}$$

Solution set:

$$\boxed{\{x : -3 \leq x < \frac{7}{5}, x \in \mathbb{R}\}}$$

Number line representation:

Closed circle at -3, open circle at 7/5, shaded region between them.

4.

Ans:

From the number line:

- Closed dot at $-3 \Rightarrow -3$ is included
- Open dot at $2 \Rightarrow 2$ is not included

- Shaded region is between -3 and 2

So, the solution set is:

$$\{x : x \in \mathbb{R}, -3 \leq x < 2\}$$

Correct answer: (a)

5.

Ans:

Solve the inequation:

$$x - 3 \leq -5$$

Add 3 to both sides:

$$x \leq -2$$

Since $x \in \mathbb{R}$,

Solution set:

$$\{x : x \leq -2, x \in \mathbb{R}\}$$

Correct answer: (b)

6.

Ans:

Solve the compound inequation:

$$2y - 6 < y + 2 \leq 2y, \quad y \in \mathbb{N}$$

Step 1:

$$2y - 6 < y + 2 \Rightarrow y < 8$$

Step 2:

$$y + 2 \leq 2y \Rightarrow y \geq 2$$

Combine:

$$2 \leq y < 8$$

Since $y \in \mathbb{N}$,

$$y = \{2, 3, 4, 5, 6, 7\}$$

Number line: solid points at 2, 3, 4, 5, 6, 7.

Correct option: (c)

7.

Ans:

Given inequation:

$$-8 \leq x - 7 < -4, \quad x \in I$$

Add 7 to all three parts:

$$-8 + 7 \leq x < -4 + 7$$

$$-1 \leq x < 3$$

Since $x \in I$ (integers),

$$x = \{-1, 0, 1, 2\}$$

Correct answer: (d) $\{-1, 0, 1, 2\}$